Ayush Agarwal

+1 408-375-8756 | agarw534@purdue.edu | LinkedIn | GitHub | Portfolio

Education

Purdue University, College of Science — West Lafayette, IN

B.S. Computer Science — *Expected May 2028*

- Coursework: Proof-based Linear Algebra, Discrete Mathematics, Data Structures & Algorithms, Systems Programming in C, Computer Architecture, Probability, Competitive Programming, Object-Oriented Programming in Java
- Awards: AIME Qualifier, AMC Distinction

Experience

AI Engineering Intern — May 2026 – Aug 2026

Skan AI — *Menlo Park, CA*

- Designed and built an agentic pipeline converting raw step-level process logs into executable workflows via automated event grouping and dynamic tool selection, validated across 6+ enterprise processes for customers like Fineos.
- Built a generate-validate-repair loop for browser automation, pairing static checks and headless execution on live pages with an LLM judge over screenshots; reached a 95% pass rate over 5 runs of a 6-scenario benchmark I wrote, caching passes as context
- Built a synthetic data pipeline yielding 8,000 labeled task recordings across 210 process variants (∼800 events each), expanding process specs into action plans and running them on real web and desktop apps via a reusable catalogue of OS-level drivers
- Made simulated sessions behave like real operators, injecting 17 distinct noise patterns into every run and wrapping each action in per-persona typing, mouse, and reading models so pacing, hesitation, and error rates vary by participant

Research Assistant — Prof. Tianyi Zhang — Jan 2026 – Present

Purdue University — *West Lafayette, IN*

- Characterized LLM program-repair failure modes across SWE-bench Verified, Lite, and Pro, presenting weekly findings to 3 PhD researchers, by categorizing agent trajectories on bug type, localization behavior, iteration count, cost, and token usage
- Grounded the architecture of a memory-augmented repair pipeline in 30+ surveyed papers on program repair, memory systems, and fault localization, and co-designed an agent that indexes repository-specific bug patterns and prior patch outcomes to improve later localization attempts

Software Engineering Intern — Jan 2025 – Jun 2025

MightyBot — *Palo Alto, CA*

- Shipped RepoChat, a GPT-4o codebase analysis platform answering multi-file questions over arbitrary cloned GitHub repositories, by building async retrieval over a source-file map with agentic query decomposition
- Cut end-to-end latency 60%+ by profiling the request pipeline for sequential bottlenecks and replacing them with batched async I/O and semaphore-bounded concurrency across parallel inference calls
- Improved answer relevance 30%+ in LLM-as-judge evaluations by implementing LangChain and Pydantic query decomposition that split repo-level questions into file-scoped subtasks

Software Engineering Intern — Dec 2023 – Feb 2024

Awareye — *Palo Alto, CA*

- Raised computer-vision precision 5% on a client-image test set through hyperparameter tuning, image labeling, annotation correction, and validation
- Shipped client-facing interfaces to iOS by building responsive views in Next.js and React, packaging them natively with Capacitor, and tailoring behavior in Swift

Research Assistant — Prof. Koushik Sen — Oct 2022 – Jul 2023

UC Berkeley — *Berkeley, CA*

- Supplied the training corpus for a code idiom similarity model, contributing to arxiv.org/abs/2312.15157, by building a Python data-mining pipeline that extracted, deduplicated, and clustered snippets across 100+ GitHub repositories

Projects

AgentQR | *Python, LangChain, RAG, Backtesting* | Project Lead, Boiler Quant — Jan 2026 – Present

- Leading a 5-person team building a multi-agent research system where specialized LLM agents collect market data, debate hypotheses, and surface cross-sector trading signals
- Produced strategies backtested to Sharpe ∼0.75 over 5 years of historical data by architecting an orchestration layer that runs 3 specialist agents through adversarial debate cycles, grounded in RAG pipelines over SEC filings and earnings transcripts with cross-encoder reranking

Disadus | *Next.js, React, Node.js, MongoDB* | Co-founder — Sep 2022 – Jun 2025

- Scaled a full-stack learning management system to 500+ active students across 8 schools with 40% faster page loads, via SSR optimization, lazy-loaded component splitting, and asset compression

Technical Skills

Languages: Python, C, C++, Java, TypeScript, JavaScript
AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, RAG, LLM evaluation, LLM-as-judge
Backend & Systems: FastAPI, Docker, Git, PyTest, async Python, concurrency, OpenAI API
Frontend/Mobile: React, Next.js, React Native, Swift